[INCH-POUND]
A-A-3093
November 6, 1996

COMMERCIAL ITEM DESCRIPTION

SANDER, ELECTRIC, PORTABLE, OSCILLATING PAD

The General Services Administration has authorized the use of this Commercial Item Description for procurement of type III sanders specified in OO-S-90C for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description covers portable, electric sanders with oscillating pads for finish sanding of wood and other materials.

2. CLASSIFICATION.

2.1 Classification. Not applicable.

3. SALIENT CHARACTERISTICS.

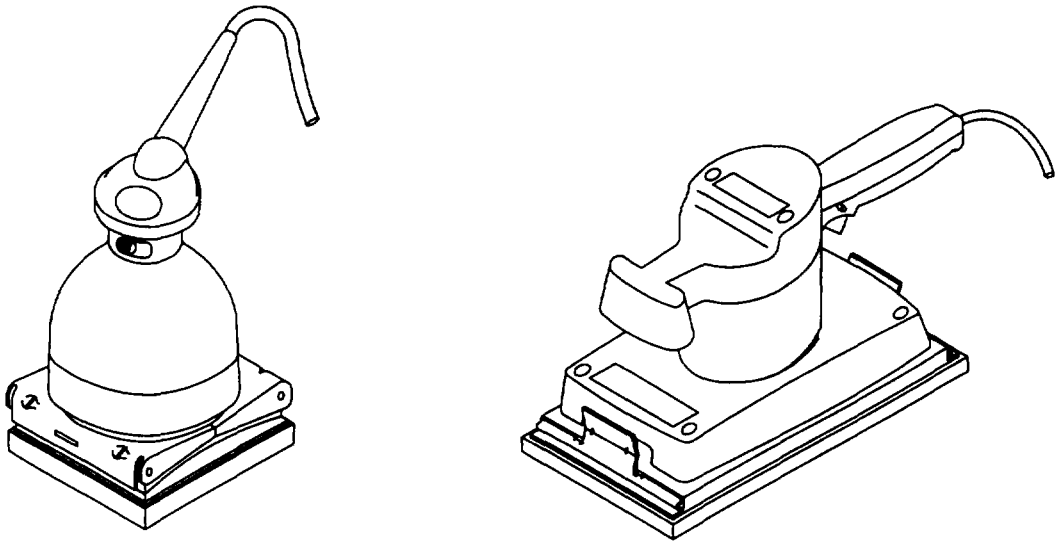
3.1 Design. The oscillating pad sanders shall consist of housings, electric motor, switch, wiring, gears, shafts, bearings, motion transfer mechanism, backing pad, abrasive holding device, handles, and standard equipment with a design to provide convenient and safe operation. The sander shall have no-load orbital motion of not less than 12,000 orbits per minute for the 1/4-sheet size sander and 9,000 orbits per minute for the 1/4-sheet size sander. The sander shall conform to the requirements of Underwriters' Laboratories UL 45 for electrical tools; safety devices, guards, and ventilating openings shall be within the acceptable safety requirements of UL 45. Convenient access shall be provided to areas where the manufacturer recommends periodic lubrication or lubricant replacement. All parts subject to wear, breakage, or distortion shall be easily accessible for adjustment and repair. All parts from the same manufacturer shall be interchangeable without modifiable adjustment. The 1/4-sheet sanders shall be similar to style 1 shown in Figure 1 and the 1/3-sheet and 1/2-sheet sanders shall be similar to style 2 shown in figure 1 and conform to the requirements specified in Table I.

3.2 Mechanical features. The castings and housings shall be free of defects or mismatching and made of material suitable to withstand the stresses imposed by normal usage. Fastening devices shall be installed with a means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be permanently deformed. All bearings shall be precision quality, commercially available ball or roller bearings permanently lubricated with dust-proof seals. Gears and shafts shall be fully enclosed and of suitable material designed and supported to efficiently transmit imposed loads.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center, Kansas City, MO 64131.

Distribution Statement A. Approved for public release; distribution is unlimited. FSC 5130

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Style 1 Style 2

Figure 1. Sanders, oscillating pad.

TABLE I. Sanders, oscillating pad

Size	Weight less cable (max.) (Pounds)	Test weight (Pounds)	Abrasive area (min.) (Square inches)	Amplitude of oscillations	
				Minimum (Inch)	Maximum (Inch)
1/4 Sheet	6	10	18	0.078	0.375
1/3 Sheet	7-1/2	10	25	0.125	0.500
1/2 Sheet	8-1/2	10	40	0.125	0.500

3.2.1 Motion transfer mechanism. When specified (see 7.2), style 2 sanders shall have a motion transfer mechanism to transfer motor rotation to orbital motion of the sanding pad. Moving elements of the mechanism shall be mounted in ball bearings designed to withstand the imposed radial thrust loads. Motive power shall be transmitted to the pad by belt, gear train, cam, direct drive, or other means.

3.2.2 Abrasive holding device. Each sander shall be furnished with a permanently attached abrasive holding device designed to quickly remove and securely replace abrasive paper.

3.2.3 Handles. The sanders shall be designed for two-hand operation and shall have a straight or pistol grip handle and an auxiliary knob handle designed for a comfortable grip and ease of operation. The sanders designed for one-hand operation shall have a functional single grip.

3.2.4 Standard equipment. The sanders shall be equipped with a backing pad of felt, rubber, or other durable resilient material attached to a metal plate.

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3.3 Electrical features. The sander's motor shall be of a universal series wound or compensated series wound designed to operate on either 115 volts direct current (DC) or 115 (± 6 percent), single phase, 25 to 60 cycles alternating current (AC). The motor armature shall be dynamically balanced to ensure freedom from destructive vibrations. The motor commutator may be of V ring and sheet mica or of molded construction. The mica shall be undercut between the commutator segments. Molded commutators shall be of such material that it will withstand maximum rotational forces and be immune to deformation from temperature encountered in service. The wiring, flexible cord, and attachment plug shall be in accordance with UL 45, as applicable. The cord shall have an attachment plug cap, and unless otherwise specified (see 7.2), shall have a minimum length of 6 feet measured from point of entry into the sander. The attachment plug cap shall conform to UL 498 requirements for 2-pole, 3-wire grounding devices rated 15 amperes at 125 volts, unless otherwise specified (see 7.2). The sander shall have a safe and conveniently located trigger switch with a current rating of not less than the rated load current of the sander. The switch shall have a lock for holding in the "ON" position provided that actuation and turnoff can be accomplished by the same finger(s) that actuate the switch.

3.4 Load testing. When required to be tested, the sander shall be operated for a minimum of 30 minutes at no-load. During the test, the sander shall be checked for switch operation or any loss of lubricant. After completing the no-load test, the sander shall be operated at the rated load. When operating temperatures are constant, the sander shall be operated at the rated load for a minimum of one hour. The sander shall then be tested for temperature rise, current input, and dielectric strength and insulation resistance in accordance with the applicable requirements of UL 45. When the rated load test is completed, operational components (including motor bearings and gears) shall be examined for wear, insecure fastening, and evidence of leakage. Failure to meet any of the above tests shall be cause for rejection.

3.5 Additional requirements.

3.5.1 Electromagnetic interference suppression. When specified (see 7.2), the sander shall be electromagnetic interference suppressed in accordance with the requirements of MIL-STD-461.

3.5.2 Fungus proofing. When specified (see 7.2), the electrical circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and water resistant, or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose function would be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied to give a minimum dry film thickness of approximately 0.001 inch to the component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.6 Handbook of instruction. The manufacturer shall furnish one copy of the regular instruction book and parts list with each sander. The handbook shall include all illustrations, instructions, and necessary data for safe operation, maintenance, adjustment, and repair. The parts list shall include the manufacturer's part number and ordering instructions.

3.7 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.8 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material

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workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified requirements. For product characteristics and requirements which are not contractually specified, the provided product shall conform to the producer's own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the producer. If the product is not commercially offered by the producer, the best commercial practices shall apply. The Government reserves the right to require proof of such conformances prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

6. PACKAGING.

6.1 Packaging. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Specifications and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

Federal Acquisition Regulations (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

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DoD Standardization Documents: Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 1911-5094.

UL 45 and UL 498: Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) When applicable, the appropriate NSN(s).
(c) When applicable, the specific styles, as specified in the Commercial Item Description.
(d) If bid samples are required, the number of samples required, and the subjective inspection requirements.
(e) If first article samples are required and the number of samples required.
(f) If lot sampling inspection is required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The Inspection Level shall be S-2 with an Acceptable Quality Level of 4.0, expressed in terms of defects per hundred units.

(g) Cord length, if different (see 3.3).
(h) Attachment plug conformance to UL 498, if required (see 3.3).
(i) Electromagnetic interference suppression, if required (see 3.5.1).
(j) Moisture and fungus treatment, if required (see 3.5.2).

7.3 National Stock Numbers (NSNs). The following list of NSNs corresponds to the sizes specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

NSN	MODEL
5130-00-990-5238	1/4 sheet
5130-00-596-9727	1/3 sheet
5130-00-409-8653	1/2 sheet

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions or notices to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

GSA - FSS

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